

Africast-Tidy Time Series & Forecasting Using R

<https://workshop.f4sg.org/africast/>



Project lead and Instructors



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Instructors



Abiodun Egwuenu

Medical doctor, field epidemiologist & researcher specialising in antimicrobial resistance (AMR). National lead for Nigeria's mpox response (2022/23). PhD candidate at Charité – Universitätsmedizin Berlin.



Harsha Halgamuwe Hewage

Researcher, Data Lab for Social Good, Cardiff Business School. Works on forecasting & decision support under uncertainty for public health supply chains, with JSI, USAID and the HISP Centre, Oslo. Delivers F4SG training and researches time series foundation models.



Mustafa Aslan

Third-year PhD researcher, Data Lab for Social Good, Cardiff University. Researches hospital discharge delays and patient flow, combining probabilistic forecasting, sequential optimisation and ML-based causal inference. Developer of the `peshbeen` forecasting package. Part of the team placing joint first in the 2026 NHS SPHERE-PPL forecasting challenge.

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Facilitator

■ Abiodun Egwuenu

Planning team

- Mkajuma Charity
- Kennedy Mung'are
- José Thiéry Hagbe

Mentors

- Samson Oluwafemi
- Oluchukwu Aghadi
- Lumumba Victor
- Adamu SAIDU Saleh
- Akoeugnigan Idelphonse SODE
- AKOUNDA Badjibassa

Coordinator

- Henry Kissinger Ochieng

Assumptions

- This is not an introduction to R. We assume you are broadly comfortable with R code, the RStudio environment and the tidyverse.
- This is not a statistics course. We assume you are familiar with concepts such as the mean, standard deviation, quantiles, regression, normal distribution, likelihood, etc.
- This is not a theory course. We are not going to derive anything. We will teach you time series and forecasting tools, when to use them, and how to use them most effectively.

Key reference

Hyndman, R. J. & Athanasopoulos, G. (2021)
***Forecasting: principles and practice*, 3rd ed.**

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Hyndman, R. J. & Athanasopoulos, G. (2021)
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[OTexts.org/fpp3/](https://otexts.org/fpp3/)

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Hyndman, R. J. & Athanasopoulos, G. (2021)
***Forecasting: principles and practice*, 3rd ed.**

[OTexts.org/fpp3/](https://otexts.org/fpp3/)

- Free and online
- Data sets in associated R package
- R code for examples

Reference - Recommended

Kolassa S., Rostami-Tabar B., & Siemsen, E. (2023)
<https://dfep.netlify.app/>, 1st ed.

Reference - Recommended

Kolassa S., Rostami-Tabar B., & Siemsen, E. (2023)
<https://dfep.netlify.app/>, 1st ed.

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- Free and online
- Not an in-depth technical book
- Mindset behind forecasting
- Overview of forecasting methods and processes

International Institute of Forecasters (IIF)

- Certificate will be provided by IIF. You need to attend all lectures and exercise sessions to get the certificate.
- a nonprofit organisation founded in 1982, is dedicated to developing and furthering the generation, distribution, and use of knowledge on forecasting

Forecasting for Social Good committee

- Follow us on LinkedIn
- Follow us on Twitter
- Use #AFRICAST, #F4SG if you post on social media about the workshop

Poll: How proficient are you in using R?

- 1 Guru: The R core team come to me for advice.
- 2 Expert: I have written several packages on CRAN.
- 3 Skilled: I use it regularly and it is an important part of my job.
- 4 Comfortable: I use it often and am comfortable with the tool.
- 5 User: I use it sometimes, but I am often searching around for the right function.
- 6 Learner: I have used it a few times.
- 7 Beginner: I've managed to download and install it.
- 8 Unknown: Why are you speaking like a pirate?

Poll: How experienced are you in forecasting

- 1 Guru: I wrote the book, done it for decades, now I do the conference circuit.
- 2 Expert: It has been my full time job for more than a decade.
- 3 Skilled: I have been doing it for years.
- 4 Comfortable: I understand it and have done it.
- 5 Learner: I am still learning.
- 6 Beginner: I have heard of it and would like to learn more.
- 7 Unknown: What is forecasting? Is that what the weather people do?

Install required packages

```
install.packages(c(  
  "tidyverse",  
  "fpp3",  
  "GGally",  
  "ggtime",  
  "mixtime",  
  "astsa"  
))
```

Approximate outline

Session	Topic	Chapter
1	1. Basics of time series and data structures	2
1	2. Time series patterns and basic graphics	2
2	3. Computing and visualising features	4
2	4. Basic modelling / forecasting	1,3,5
3	5. Forecasting with regression	7,10
3	6. ARIMA models	9
4	7. Exponential smoothing	8
4	8. Basic training and test accuracy	5
5	9. Residual diagnostics and cross validation	5
5	10. Time series foundation models	–

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Schedule for each session (UTC+3)

- Start at 10:00 a.m.
- Lecture 1: 10:00 - 11:30
- Break: 11:30 - 11:45
- Lecture 2: 11:45 - 13:15
- Exercises: 14:00 - 17:00 (on your own)
- Mentorship: 17:00 - 18:00

Access materials

- AFRICAST workshop website
- Mentorship sessions and Q/A
- Google Classroom